Further Strategies — problem 14.24

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Let $(1+\sqrt{2})^n = x_n + y_n\sqrt{2}$, where x_n, y_n are integers. Prove that
(a) $x_n^2 - 2y_n^2 = (-1)^n$; (b) $x_{n+1} = x_n + 2y_n$, $y_n = x_n + y_n$.